

Lecture 5: Measure of shape

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Learning Outcomes

- Define and compute the coefficient of variation.
- Explain the nature of skewness and kurtosis in a distribution
- Interpret the coefficients of skewness and kurtosis.

Coefficient of Variation CV

- Provides a relative measure of the dispersion in data
- (CV) for a data set defined as the ratio of the standard deviation to the mean:

$$CV = \begin{cases} \frac{\sigma}{\mu} & \text{for population} \\ \frac{s}{\bar{x}} & \text{for sample} \end{cases}$$

- It shows the extent of variability in relation to mean of the population.
- It is a normalized measure of dispersion of a probability distribution or frequency distribution.

Example

- ◇ **Example:** Rates of return over the past 6 years for two mutual funds are shown below.

Fund A	8.3	-6.0	18.9	-5.7	23.6	20
Fund B	12	-4.8	6.4	10.2	25.3	1.4

- **Problem:** Which fund has higher risk?
- **Solution:** The CV's are given as:

$$\bar{x}_A = 9.85, \bar{x}_B = 8.42, s_A = 13.19, s_B = 10.29$$

$$CV_A = \frac{13.19}{9.85} = 1.34, CV_B = \frac{10.29}{8.42} = 1.22$$

- There is more variability in Fund A as compared to Fund B. Therefore, Fund A is riskier.

Properties of Coefficient of Variation

Properties

- All values are used in the calculation.
- It is only defined for ratio level of data.
- The actual value of the CV is independent of the unit in which the measurement has been taken, so it is a dimensionless number.

For comparison between data sets with different units or widely different means, one should use the coefficient of variation instead of the standard deviation.

Skewness

- Skewness refers to the asymmetry or lack of symmetry in the shape of a frequency distribution.
- In a symmetrical distribution, the Mean, Median and Mode are equal to each other.
- In asymmetrical distribution, the *mean $\neq$ mode $\neq$ median*
- ◇ **Positively Skewed Distribution:** Mean is greater than the mode and the median lies somewhere in between mean and mode.
- ◇ **Negatively Skewed Distribution:** Mode is greater than the mean and the median lies in between mean and mode.

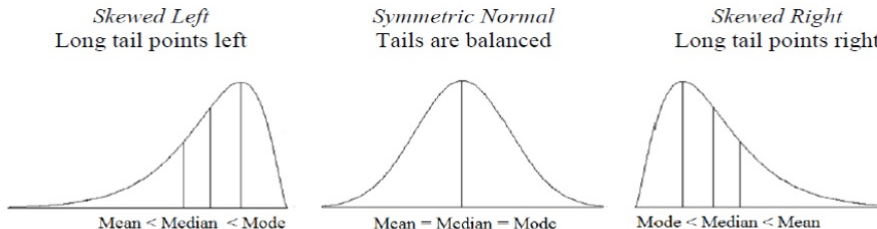


Figure 1. Sketches showing general position of mean, median, and mode in a population.

Measures of Skewness [1]

Karl Pearson' s coefficient of Skewness S_k

- **Formula;**

$$S_k = \frac{\text{mean} - \text{mode}}{\sigma} = \frac{3(\text{mean} - \text{median})}{\sigma}$$

- **Interpretation;** if

- ◇ $S_k = 0$ the distribution of the data is symmetrical
- ◇ $S_k > 0$ The distribution of the data is positively skewed
- ◇ $S_k < 0$ The distribution of the data is negatively skewed

Example

- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- Arranging the data in ascending order gives: 14.0, 15.0, 15.0, 16.0, 17.0
- ◇ **Solution:**

$$\bar{x} = 15.4, s = 1.14, mode = 15$$

$$S_k = \frac{15.4 - 15}{1.14} = 0.35$$

- ◇ $S_k > 0$ The distribution of the bonuses is positively skewed.

Measures of Skewness [2]

Bowleys's S_Q

- Method is based on quartiles
- **Formula;**

$$S_Q = \frac{Q_3 - 2 \times Q_2 + Q_1}{Q_3 - Q_1}$$

- $Q_1 = 1^{st} \text{ quartile}$, $Q_2 = 2^{nd} \text{ quartile}$, $Q_3 = 3^{rd} \text{ quartile}$
- **Interpretation;** if
 - ◇ $S_Q = 0$ the distribution of the data is symmetrical
 - ◇ $S_Q > 0$ The distribution of the data is positively skewed
 - ◇ $S_Q < 0$ The distribution of the data is negatively skewed

Example

- ◇ **Example:** Bonus example
(\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- Arranging the data in ascending order gives:
14.0, 15.0, 15.0, 16.0, 17.0
- ◇ **Solution:**
 - Q_1 position is $n/4 = 5/4 = 1.25 \simeq 1^{st} \text{ position} = 14.0$.
 - Q_2 position is $n/2 = 5/2 = 2.5 \simeq 3^{rd} \text{ position} = 15.0$
 - Q_3 position is $\frac{3n}{4} = \frac{3 \times 5}{4} = 3.75 \simeq 4^{th} \text{ position} = 16.0$.
 - $S_Q = \frac{16 - 2 \times 15 + 14}{16 - 14} = 0$
- ◇ $S_k = 0$ The distribution of the bonuses is symmetrical.

Measures of Skewness [3]

Kelly's S_P

- Method is based on percentiles
- **Formula;**

$$S_P = \frac{P_{90} - 2 \times P_{50} + P_{10}}{P_{90} - P_{10}}$$

- $P_{90} = 90^{th}$ percentile, $P_{50} = 50^{th}$ percentile, $P_{10} = 10^{th}$ percentile
- **Interpretation;** if
 - ◇ $S_P = 0$ the distribution of the data is symmetrical
 - ◇ $S_P > 0$ The distribution of the data is positively skewed
 - ◇ $S_P < 0$ The distribution of the data is negatively skewed

Example

- ◇ **Example:** Bonus example
(\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- Arranging the data in ascending order gives:
14.0, 15.0, 15.0, 16.0, 17.0
- ◇ **Solution:**
 - P_{90} position is $0.9N = 0.9 \times 5 = 4.5 \simeq 5^{th} position = 17$.
 - P_{50} position is $0.5N = 0.5 \times 5 = 2.5 \simeq 3^{rd} position = 15$
 - P_{10} position is $0.1N = 0.1 \times 5 = 0.5 \simeq 1^{st} position = 14$
 - $S_P = \frac{17 - 2 \times 15 + 14}{17 - 14} = \frac{1}{3} = 0.333$
- ◇ $S_P > 0$ The distribution of the bonuses is positively skewed.

Measures of Skewness [4]

Moments Coefficient of Skewness β_1

- Method is based on moments μ_r

- **Formula;**

$$\beta_1 = \frac{\mu_3}{\mu_2^3}$$

- $\mu_3 = \frac{1}{N} \sum_{i=1}^N (x - \mu)^3, \mu_2 = \frac{1}{N} \sum_{i=1}^N (x - \mu)^2$

- **Interpretation;** if

- ◇ $\mu_3 = 0$ the distribution of the data is symmetrical
- ◇ $\mu_3 > 0$ The distribution of the data is positively skewed
- ◇ $\mu_3 < 0$ The distribution of the data is negatively skewed

Example

- ◇ **Example:** Bonus example
(\$000) : 14.0, 15.0, 17.0, 16.0, 15.0

- ◇ **Solution:**

x	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^3$
14.00	-1.40	1.96	-2.74
15.00	-0.40	0.16	-0.06
15.00	-0.40	0.16	-0.06
16.00	0.60	0.36	0.22
17.00	1.60	2.56	4.10
Total	0.00	5.20	1.44

Example-continuation

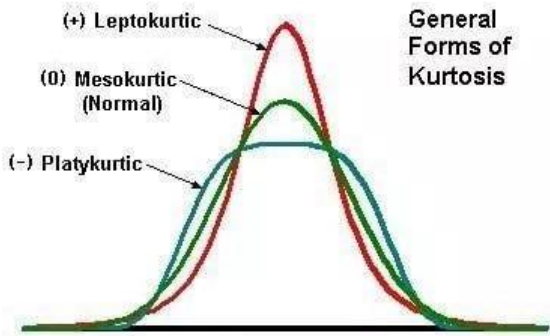
Moments Coefficient of Skewness β_1

$$\beta_1 = \frac{\mu_3}{\mu_2^3}$$

- $\mu_3 = \frac{1.44}{5} = 0.29, \mu_2 = \frac{5.2}{5} = 1.04$
- $\beta_1 = \frac{0.29}{1.04^3} = 0.258$
- **Interpretation;**
- ◇ $\mu_3 > 0$ The distribution of the data is positively skewed

Kurtosis β_2

- kurtosis refers to the degree of flatness or peakedness of a curve
- The kurtosis of a distributions is defined by
 - ◇ **Mesokurtic** The normal curve $\beta_2 = 3$.
 - ◇ **Leptokurtic distributions** : is one that has kurtosis greater than a mesokurtic distribution $\beta_2 > 3$.
 - ◇ **Platykurtic distributions** : the curve is less peaked than normal curve $\beta_2 < 3$



Measures of Kurtosis

Moments Coefficient of Kurtosis β_2

- Method is based on moments μ_r

- **Formula;**

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

- $\mu_4 = \frac{1}{N} \sum_{i=1}^N (x - \mu)^4, \mu_2 = \frac{1}{N} \sum_{i=1}^N (x - \mu)^2$

- **Interpretation;** if

- ◇ $\beta_2 = 3$ the distribution of the data is Mesokurtic.
- ◇ $\beta_2 > 3$ The distribution of the data is Leptokurtic
- ◇ $\beta_2 < 3$ The distribution of the data is Platykurtic

Example

- ◇ **Example:** Bonus example
(\$000) : 14.0, 15.0, 17.0, 16.0, 15.0

- ◇ **Solution:**

x	$(x - \mu)$	$(x - \mu)^2$	$(x - \mu)^3$	$(x - \mu)^4$
14.00	-1.40	1.96	3.84	-2.74
15.00	-0.40	0.16	-0.06	0.03
15.00	-0.40	0.16	-0.06	0.03
16.00	0.60	0.36	0.22	0.13
17.00	1.60	2.56	4.10	6.55
Total	0.00	5.20	1.44	10.58

Example-continuation

Moments Coefficient of Kurtosis β_2

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

- $\mu_4 = \frac{10.58}{5} = 2.116, \mu_2 = \frac{5.2}{5} = 1.04$
- $\beta_2 = \frac{2.116}{1.04^2} = 1.956$
- **Interpretation;**
- ◇ $\beta_2 < 3$ The distribution of the data is platykurtic

THANK YOU

QUESTIONS